

PHASE 0 COMPLETION REPORT

A. Phase

PHASE 0 – FOUNDATION / PROJECT BOOTSTRAP

B. Status

COMPLETED

C. Project Architecture

The architecture and layer boundaries specified in Section 5 & 7 of `AL-AZHARI-LIBRARY-FRONTEND-IMPLEMENTATION-PLAN.pdf` were created:

```

src/
├─ main.ts
├─ index.html          (lang="ar" dir="rtl" default, font preconnects)
├─ styles.scss         (global styles: tokens -> base -> themes)
├─ environments/
│   ├─ environment.model.ts    (strictly typed, non-secret environment interface)
│   ├─ environment.ts         (development config: apiBaseUrl = '/api/v1')
│   └─ environment.production.ts
├─ app/
│   ├─ app.config.ts         (router, zone, http interceptors, GlobalErrorHandler)
│   ├─ app.routes.ts         (top-level lazy route groups foundation)
│   ├─ app.component.ts      (root component initializing LocaleService & direction)
│   ├─ app.component.html    (accessible skip-link & <router-outlet />)
│   └─ app.component.scss
│
│   └─ core/                 (network, auth, errors, storage, i18n, config)
│       ├─ a11y/
│       │   └─ a11y.service.ts
│       ├─ analytics/
│       │   └─ analytics.service.ts (event facade with no-op default)
│       ├─ api/
│       │   ├─ base/
│       │   │   └─ api-response.model.ts (ApiResponse<T>, ApiErrorResponse, PaginationMeta)
│       │   ├─ dto/
│       │   ├─ mappers/
│       │   ├─ public/
│       │   ├─ commerce/
│       │   ├─ services/
│       │   ├─ account/
│       │   └─ admin/
│       ├─ auth/
│       ├─ config/
│       │   └─ app-config.service.ts (safe environment accessor)
│       ├─ error-handling/
│       │   └─ global-error-handler.ts (re-export)
│       ├─ errors/
│       │   └─ global-error-handler.ts (SSR-safe Angular ErrorHandler)
│       ├─ http/
│       │   ├─ request-id.interceptor.ts (attaches X-Request-ID)
│       │   └─ locale.interceptor.ts    (attaches Accept-Language)
│       ├─ i18n/
│       │   ├─ direction.service.ts     (RTL/LTR signals, document dir sync)
│       │   └─ locale.service.ts        (ar/en signals, storage persistence)
│       ├─ realtime/
│       │   └─ realtime.service.ts       (IRealtimeService foundation skeleton; no connection until Phase 8)
│       ├─ seo/
│       ├─ storage/
│       │   ├─ platform.service.ts       (isBrowser/isServer, SSR-safe window/document/storage)
│       │   └─ safe-storage.ts           (SSR-safe & sandbox-safe key-value store with in-memory fallback)
│       └─ util/
│
│   └─ domain/               (stable UI domain models & pure primitives)
│       ├─ enums/
│       ├─ mappers/
│       └─ models/
│           ├─ money.model.ts            (integer minor units representation & currency formatting)
│           └─ localized-text.model.ts   ({ ar: string; en?: string } extractor)
│
│   └─ shared/               (reusable design system primitives & utilities)
│       ├─ ui/
│       ├─ forms/
│       ├─ data-display/
│       ├─ pipes/
│       └─ directives/
│
│   └─ layout/               (shell layouts)
│       ├─ storefront/
│       ├─ account/
│       └─ admin/

```

		└─ auth/	
		└─ features/	(isolated feature bundles)
		└─ storefront/	
		└─ checkout/	
		└─ services/	
		└─ auth/	
		└─ customer/	
		└─ admin/	
		└─ styles/	
		└─ tokens/	
		└─ _color.scss	(forest green, accent gold, ivory paper, semantic feedback)
		└─ _type.scss	(IBM Plex Sans Arabic + Inter typographic scale)
		└─ _space.scss	(4px base spacing scale)
		└─ _radius.scss	(sm 6px, md 10px, lg 16px, xl 24px, pill 9999px)
		└─ _shadow.scss	(border-first elevations with modern color notation)
		└─ _motion.scss	(fast 120ms, normal 200ms, slow 320ms, prefers-reduced-motion)
		└─ _z.scss	(base 0, sticky 20, dropdown 40, drawer 60, modal 80, toast 100)
		└─ _breakpoints.scss	(xs, sm, md, lg, xl, 2xl, 3xl maps & bp-up/bp-down mixins)
		└─ _containers.scss	(reading 720px, content 1120px, wide 1280px, max 1440px)
		└─ _index.scss	
		└─ base/	
		└─ _reset.scss	(logical properties: inline-size, block-size, margin-inline)
		└─ _typography.scss	
		└─ _direction.scss	(:dir rtl), :dir ltr, .ltr-island, .flip-in-rtl)
		└─ _focus.scss	(accessible :focus-visible styling)
		└─ _utilities.scss	(.visually-hidden, .skip-link, .container)
		└─ _index.scss	
		└─ themes/	
		└─ light.scss	(semantic theme mapping)

D. Dependencies Added

Pinned Dependencies in `frontend/package.json`

Package	Version	Purpose
@angular/core	19.2.19	Core Angular framework (pinned)
@angular/common	19.2.19	Common directives, pipes, platform utilities
@angular/router	19.2.19	Standalone router
@angular/forms	19.2.19	Typed reactive forms
@angular/cdk	19.2.19	A11y, LiveAnnouncer, overlays, breakpoints
rxjs	7.8.2	Reactive streams & HTTP pipeline
@jsverse/transloco	8.4.0	Runtime scoped i18n dictionary loader
socket.io-client	4.8.1	Realtime client library (skeleton foundation only)
lucide-angular / @lucide/angular	1.0.0 / 1.52.0	Angular-compatible Lucide icon library
angular-eslint	19.8.1	Flat config ESLint support for Angular 19
eslint-plugin-boundaries	5.0.1	Architecture dependency boundary enforcement
stylelint	16.15.0	SCSS linting
stylelint-use-logical-spec	5.0.1	Strict RTL logical property enforcement
typescript	5.7.3	Pinned TypeScript compiler

Deliberately NOT Installed (Prohibited by Plan & Architecture)

- **✗ Angular Material visual components** (violates custom design system)
- **✗ PrimeNG / Bootstrap / Tailwind runtime** (violates SCSS + CSS variables design system)
- **✗ NgRx** (Angular signals + RxJS injectable stores specified by plan)
- **✗ Lodash** (native ES2022 used)
- **✗ Moment / Day.js** (native Intl API specified by plan)
- **✗ Money libraries** (integer minor units domain model implemented)

E. Configuration Files

File	Status	Action & Rationale
frontend/package.json	Modified	Pinned all Angular packages to 19.2.19 , added angular-eslint@19.8.1 .
frontend/tsconfig.json	Verified	Enforces strict: true , noUncheckedIndexedAccess: true , exactOptionalPropertyTypes: true . No bypasses.
frontend/angular.json	Modified	Configured optimization.fonts: false to allow offline/CI compilation without network timeouts.
frontend/.stylelintrc.json	Modified	Configured logical properties enforcement (liberty/use-logical-spec: "always"), modern color function notation, font-family case exemptions.
frontend/eslint.config.mjs	Verified	Enforces strict boundaries between core , domain , shared , layout , and isolated features .
frontend/src/index.html	Modified	Configured lang="ar" dir="rtl" , Google Fonts preconnects for IBM Plex Sans Arabic and Inter, Azhari title.
frontend/src/environments/environment.ts	Verified	Non-secret config only (apiBaseUrl = '/api/v1' , feature flags).
frontend/src/environments/environment.production.ts	Verified	Non-secret production config (apiBaseUrl = '/api/v1').
docs/api-contract-notes.md	Created	Gating contract verifications against backend Postman collection and source code.
docs/design-website-audit.md	Created	Gap A resolution auditing all 34 mockup folders in design-website/ .

F. Architecture Rules Enforced

- Core Boundaries:** core/ does not import anything from features/ , layout/ , or shared/ .
- Domain Boundaries:** domain/ contains only pure models and functions; depends only on low-level primitives.
- Shared Boundaries:** shared/ does not depend on features/ .
- Feature Isolation:** features/ is prohibited by ESLint from importing features/ .
- Admin Isolation:** features/admin/ is isolated from storefront and customer features.
- SSR Safety:** PlatformService and SafeStorage encapsulate all DOM, window, navigator, and storage access.
- RTL / Logical CSS:** Enforced by Stylelint using stylelint-use-logical-spec .

G. Validation Results

All checks were executed in the repository and passed:

npm install	PASS (dependencies installed and locked)
build	PASS (Application bundle generation complete in 2.38s)
TypeScript	PASS (tsc --noEmit exited with code 0)
lint (ESLint)	PASS (All files pass linting)
lint:style	PASS (stylelint "src/**/*.scss" exited with code 0)
tests (Karma)	PASS (13 of 13 tests SUCCESS in ChromeHeadless)

Karma Test Execution Summary

- AppComponent : creates app, initializes Arabic title, verifies skip link and main content area.
- PlatformService : correctly distinguishes browser vs server platforms; verifies safe null fallback on server.
- SafeStorage : safely stores, retrieves, and clears keys with in-memory fallback.
- LocaleService : defaults to Arabic with RTL direction; switches to English with LTR direction.
- Money Model : enforces integer minor units; validates major/minor conversions; formats currency in Arabic (₭.؄) and English (EGP).

H. Security Check

CONFIRMED:

- No secrets added to frontend configuration.
- No JWT secrets or signing keys present.
- No database credentials or MongoDB URIs present.
- No private API keys or Cloudinary secrets present.
- Frontend uses apiBaseUrl = '/api/v1' for relative same-origin deployment.
- Signed upload parameters and auth tokens are received dynamically from backend at runtime.

I. Files Changed

Created

- docs/api-contract-notes.md
- docs/design-website-audit.md
- frontend/src/app/core/storage/platform.service.ts
- frontend/src/app/core/storage/platform.service.spec.ts
- frontend/src/app/core/storage/safe-storage.ts
- frontend/src/app/core/storage/safe-storage.spec.ts
- frontend/src/app/core/config/app-config.service.ts
- frontend/src/app/core/i18n/direction.service.ts
- frontend/src/app/core/i18n/locale.service.ts
- frontend/src/app/core/i18n/locale.service.spec.ts
- frontend/src/app/core/http/request-id.interceptor.ts
- frontend/src/app/core/http/locale.interceptor.ts
- frontend/src/app/core/errors/global-error-handler.ts
- frontend/src/app/core/error-handling/global-error-handler.ts

- frontend/src/app/core/realtime/realtime.service.ts
- frontend/src/app/core/analytics/analytics.service.ts
- frontend/src/app/core/a11y/a11y.service.ts
- frontend/src/app/core/api/base/api-response.model.ts
- frontend/src/app/domain/models/money.model.ts
- frontend/src/app/domain/models/money.model.spec.ts
- frontend/src/app/domain/models/localized-text.model.ts
- frontend/src/styles/tokens/_radius.scss
- frontend/src/styles/tokens/_shadow.scss
- frontend/src/styles/tokens/_motion.scss
- frontend/src/styles/tokens/_z.scss
- frontend/src/styles/tokens/_breakpoints.scss
- frontend/src/styles/tokens/_containers.scss
- frontend/src/styles/tokens/_index.scss
- frontend/src/styles/base/_reset.scss
- frontend/src/styles/base/_typography.scss
- frontend/src/styles/base/_direction.scss
- frontend/src/styles/base/_focus.scss
- frontend/src/styles/base/_utilities.scss
- frontend/src/styles/base/_index.scss
- frontend/src/styles/themes/light.scss

Modified

- frontend/angular.json
- frontend/package.json
- frontend/.stylelintrc.json
- frontend/src/index.html
- frontend/src/styles.scss
- frontend/src/styles/tokens/_color.scss
- frontend/src/styles/tokens/_type.scss
- frontend/src/app/app.config.ts
- frontend/src/app/app.component.ts
- frontend/src/app/app.component.html
- frontend/src/app/app.component.spec.ts

Deleted

- None. (Backend in `Back-end/` was left completely untouched).

J. Deferred Work (Outside Phase 0 Scope)

- **Phase 1 Primitives:** Buttons, Inputs, Cards, Badges, Modals, StatusPresentation, Toast, and living stylebook.
- **Phase 2 Infrastructure:** Typed API endpoint services, DTO mappers, AuthStore, full Transloco dictionaries, SeoService.
- **Phase 3 Authentication:** Login, registration, password recovery, verification UI.
- **Phase 4+ Commerce & Management:** Storefront, catalog, cart, checkout, orders, services, pre-orders, and admin dashboard.

K. Blockers / Warnings

- **Blockers:** None. All Phase 0 blockers, architectural decisions, and contract checks have been resolved.
- **Warnings:** None. All builds, lints, styles, and unit tests pass with zero warnings or errors.
- **Deferred:** All feature and page development deferred to subsequent phases in strict sequence.

Awaiting explicit authorization to proceed to Phase 1 (Design system and shared primitives).